

1. Q-Learning with DQN

A robot uses a Deep Q-Network (DQN) for navigating a grid. The network has four actions: up, down, left, and right, each giving a reward of +10 if it reaches the goal state or -1 otherwise. The robot starts from the top-left cell (0,0) and the goal is located at the bottom-right cell (4,4).

- **Problem:** If the discount factor is 0.9 and learning rate is 0.1, calculate the Q-value update for taking an action that moves from (1,1) to (1,2) with the current Q-value at 5. Show the Q-value update calculation using the Bellman equation.

2. Double DQN Update

In Double DQN, two networks, the primary network, and the target network, are used to avoid overestimation. Consider a state where the primary network outputs the Q-values $Q_{up}=5, Q_{down}=7, Q_{left}=4, Q_{right}=6$. The target network outputs $Q_{up}=5, Q_{down}=6, Q_{left}=4, Q_{right}=7$.

- **Problem:** Calculate the Q-value update if the agent selects the action “down” in the current state. Assume the discount factor is 0.95 and the reward received is 2.

3. Actor-Critic Policy Update

In the Actor-Critic method, an agent gets a reward of +5 for reaching a target and -2 for each step taken. Given that the Critic network provides a value estimate of the current state as 3, and the policy network output for the selected action is 0.6.

- **Problem:** Calculate the Actor-Critic gradient update assuming the discount factor is 0.8, and that the agent takes two steps to reach the target.

4. Prioritized Experience Replay

In Prioritized Experience Replay, each experience is assigned a priority score based on the TD error. Assume four experiences have TD errors of 0.3, 0.5, 0.2, 0.8. Calculate the probability of selecting each experience when the priority exponent $\alpha=0.6$.

- **Problem:** Find the normalized probabilities of selection for each experience based on their priority scores.

5. Replay Buffer with Experience Sampling

A Replay Buffer can hold up to 1000 experiences. After every 10 episodes, the agent adds 50 new experiences. If the buffer starts empty, calculate how many experiences are in the buffer after 200 episodes.

- **Problem:** At which episode will the buffer become full, and how many experiences will it hold thereafter?

6. In a Deep Q-Network (DQN) setup, an agent needs to balance between exploration (trying new actions) and exploitation (choosing the best-known actions).

- **Problem:** Explain the importance of the exploration-exploitation tradeoff in DQN. What techniques can be used to maintain this balance, and how do they impact learning?

7. Role of Target Network in DQN

DQN uses a separate target network to stabilize training.

- **Problem:** Describe why the target network is necessary in DQN. How would the learning process be affected if both the target and primary networks were updated simultaneously?

8. One of the limitations of DQN is that it tends to overestimate Q-values, leading to suboptimal policies in some cases.

- **Problem:** Explain how Double DQN mitigates the problem of Q-value overestimation. What is the main conceptual difference between DQN and Double DQN, and how does it improve the reliability of action selection?

9. In the Actor-Critic method, two networks (the Actor and the Critic) work together to optimize the policy.

- **Problem:** Describe the roles of the Actor and the Critic in this method. How does the Critic guide the Actor, and why is it advantageous to separate these functions into two networks rather than using a single Q-value network as in DQN?

10. Describe the concept of Prioritized Experience Replay and its benefits over uniform sampling. How does prioritizing certain experiences improve the learning process, and what potential drawbacks might arise from this approach?

11. The Replay Buffer stores past experiences to break correlation between consecutive samples and allow more efficient learning.

- **Problem:** Discuss why the Replay Buffer is essential in DQN and how it differs from training without one. What would happen if the Replay Buffer was too small or too large?

12. The DyNa-Q framework combines model-based and model-free learning by simulating experience using a learned model of the environment.

- **Problem:** Explain how the DyNa-Q framework utilizes simulated experiences to improve learning efficiency. What are the benefits of adding simulated planning steps, and how does it differ conceptually from pure model-free approaches?

13. The Duelling Q-Network architecture separates the Q-value function into a state value function and an action advantage function.

- **Problem:** Describe the motivation behind the Duelling Q-Network architecture. How does separating state and action values help the network focus on learning critical states more effectively?

14. In Asynchronous Advantage Actor-Critic (A3C), multiple agents operate in parallel, updating a shared global policy asynchronously.

- **Problem:** Explain how asynchronous updates in A3C help address convergence issues and lead to faster learning. What are the key advantages and challenges of asynchronous learning compared to synchronized learning methods?

15. Explain how Direct Utility Estimation works in Passive Learning. Why might this method be inefficient in environments with large state spaces or with limited observed data?

16. Describe the key concept of Temporal Difference Learning. How does TD learning update the value of a state differently from Direct Utility Estimation and Monte Carlo Methods?

17. Explain the advantages and disadvantages of Monte Carlo methods in Passive Learning. Why might it be impractical to use Monte Carlo methods in continuous or highly variable environments?

18. Compare and contrast Direct Utility Estimation, Temporal Difference Learning, and Monte Carlo Methods. In what types of environments or scenarios might each method perform best, and why?

19. Describe the significance of the exploration-exploitation tradeoff in Q-Learning. What methods can be used to balance this tradeoff, and how does the choice of method affect the agent's learning process?

20. Explain the main difference between SARSA and Q-Learning in terms of policy updates. How does SARSA's approach to on-policy learning impact the types of policies the agent learns compared to Q-Learning?

21. Discuss the advantages and disadvantages of Q-Learning versus SARSA. In what scenarios would SARSA be preferred over Q-Learning, and vice versa?